Jackson Morris

Curriculum vitae

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Simon Laufer Mathematical Sciences Institute
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Employment

Vanderbilt University — Nasvhille, TN
Postdoctoral Scholar — 2027-2029

Simon Laufer Mathematical Sciences Institute — Berkeley, CA
Postdoctoral Scholar — Fall 2026
Program: Motivic Homotopy Theory: Connections and Applications

Education

University of Washington — Seattle, WA
Ph.D. in Mathematics — 2021-2026
Advisors: Kyle Ormsby and John Palmieri
Thesis title: Cooperations in motivic homotopy theory
M.S. in Mathematics

University of Kentucky — Lexington, KY
B.S. in Mathematics *magna cum laude* — 2017-2020

Research Interests

Motivic homotopy theory, chromatic homotopy theory, hermitian K-theory, algebraic K-theory, splittings, synthetic spectra, nilpotence and periodicity, Adams spectral sequence, periodic self-maps, redshift and blueshift

Publications and Preprints

10. **Derived Galois and fracture models for cellular motivic spectra over $\mathbb{Q}_2$.**
Joint with Luca De Paulis, Andrea Figini, Marco Giustetto, and Paul Arne Østvær.
In preparation.
9. **Algebraic K-theory of Stiefel varieties.**
Joint with Sebastian Gant.
In preparation.
8. **Motivic K-theory cooperations over the rationals.**
Joint with Sarah Petersen and Elizabeth Tatum.
In preparation.
7. **A $\mathbb{C}$-motivic w_1-self-map of periodicity 1.**
Submitted. Available at https://arxiv.org/abs/2610.22651
6. **Periodic phenomena in stable motivic homotopy theory.**
Submitted. Available at https://arxiv.org/abs/2607.18165
5. **Immersions of C_2-projective spaces via K$\mathbb{R}$-theory.**
Joint with Manyi Guo, Alex Waugh, and Albert Jinghui Yang.
Submitted. Available at https://arxiv.org/abs/2604.25260.
4. **Splittings of truncated motivic Brown–Peterson cooperations algebras.**
Joint with Sarah Petersen and Elizabeth Tatum.
Submitted. Available at https://arxiv.org/abs/2509.19542.
3. **Rings of cooperations for hermitian K-theory over finite fields.**
To appear in *Algebraic & Geometric Topology*. Available at https://arxiv.org/abs/2509.02786.
2. **On the ring of cooperations for real hermitian K-theory.**
Annals of K-theory 11 (4) (2026), 633-708.
1. **Toric Double Determinantal Varieties.**
Joint with Alexander Blose, Patricia Klein, and Owen McGrath.
Communications in Algebra 49 (7) (2020), 3085-3093.

Teaching Experience

Instructor of Record

Vanderbilt University
Math 2410 – Methods of linear algebra — Spring 2027
Math 2610/5610 – Ordinary differential equations — Spring 2027

University of Washington
Math 480 – Homotopy theory — Spring 2026
Math 208 – Linear algebra — Summer 2023
Math 208 – Linear algebra — Winter 2023

Graduate Assistant

Electronic Computational Homotopy Theory (eCHT)
Research Seminar — 2025–2026
Quadratic Curve Counting — Fall 2024

Park City Math Institute (PCMI)
Experimental Math Lab — Summer 2024

University of Washington
Teaching Assistant — 2021-2026
Precalculus, Differential calculus, Multivariable calculus, Linear algebra, Introduction to proofs, Topology

Advising/Mentoring

Directed Reading Mentor — UW
Homotopy theory, Representation theory, Geometric group theory, Homological algebra — 2022–2025

Advising/Menthorship

Jason Yeung (University of Toronto) — SLMath
Program associate mentor during thematic program — Fall 2026
Spencer Catron (University of Washington) — UW
Graduate student mentor during first year — 2023-2024

Math Olympiad Judge

UW Math Hour
Judge for elementary school math competition — 2023-2024

Honors and Awards

Lee and Cozette McFarlan Travel Fellowship — UW
For travel to European Talbot in Kolding, DK. — Summer 2025

Research Assistantship — UW
For good research progress. — Summer 2025

JC Eaves Scholarship — UKY
For continued excellence in mathematics as a Kentuckian. — 2020

Carolyn Bunyan Award — UKY
For mathematics majors interested in continuing education in mathematics. — 2020

Service and Professional Activities

Refereeing and Reviewing

Algebraic and Geometric Topology — 2025-Present
ZBMath — 2026-Present

Seminar Organization

Learning Seminar — SLMath
Motivic stable stems and Galois approximations of cellular motivic categories — Fall 2026

DUBTOP Seminar — UW
Chromatic homotopy theory, stacks, Goodwillie calculus — 2024-2025

Back-To-School seminar — UW
Student lectures on fun math learned over the summer — Fall 2024

Reading seminars — UW
Algebraic K-theory, cohomology operations, motivic homotopy theory, vector bundles and topological K-theory, categorical homotopy theory, model categories — 2023-2024

Selected Presentations & Seminar Talks

Conference and Colloqium Talks

AMS Sectional: Computational Homotopy Theory — Tempe, AZ
$\mathbb{C}$-motivic periodic self-maps — November 2026

Motivic Homtopy Theory: Connections and Applications — SLMath, Berkeley, CA
Motivic K-theory cooperations over the rationals — October 2026

Young Topologists Meeting — Copenhagen, DK
Higher Witt K-theories — July 2026

Cascade Topology Seminar — Portland, OR
Higher Witt K-theories — May 2026

Joint Mathematics Meetings 2026 — Washington, D.C.
Splittings and cooperations in motivic homotopy theory — January 2026

Graduate Student Geometry & Topology Seminar — Bloomington, IN
On the ring of cooperations for real hermitian K-theory — April 2025

Reed Mathematics Colloquium — Portland, OR
Finding Fixed Points with Invariants — February 2025

Joint Mathematics Meetings 2025 — Seattle, WA
On the ring of cooperations for real hermitian K-theory — January 2025

Seminar Talks

New Mexico State University — Las Cruces, NM
$\mathbb{C}$-motivic periodic self-maps — September 2026

eCHT Research Seminar — Online
$\mathbb{C}$-motivic periodic self-maps — September 2026

University of Washington — Seattle, WA
Spectrum level splittings of truncated motivic Brown-Peterson cooperations algebras — February 2026

University of Illinois Urbana-Champaign — Champaign, IL
Splittings and cooperations in motivic homotopy theory — January 2026

University of Virginia — Charlottesville, VA
Splittings and cooperations in motivic homotopy theory — November 2025

University of Kentucky — Lexington, KY
Splittings and cooperations in motivic homotopy theory — November 2025

Duke University — Durham, NC
Splittings and cooperations in motivic homotopy theory — October 2025

University of Colorado Boulder — Boulder, CO
Spectrum level splittings of truncated motivic Brown-Peterson cooperations algebras — September 2025

University of Michigan — Ann Arbor, MI
On the ring of cooperations for real hermitian K-theory — September 2025

University of Notre Dame — South Bend, IN
Splittings and cooperations in motivic homotopy theory — September 2025

University of Washington — Seattle, WA
On the ring of cooperations for real hermitian K-theory — June 2025

Workshop Contributed Talks

Talbot	Nacogdoches, TX
$\pi_*\mathrm{THH}(\mathbb{Z}_p^{\mathrm{fil}})/p$ and $\pi_*\mathrm{THH}(\ell_p^{\mathrm{fil}})/(p, v_1)$	August 2024
European Autumn School in Topology (EAST)	Utrecht, NL
Hochschild Homology	September 2024

Reading Seminar Talks

An upper bound on the degree of symmetry of an exotic sphere	March 2026
Higher Adams differentials and hidden extensions	November 2025
Motivic twisted K-theory	April 2024
Higher real K-theories, redshift, and blueshift	May 2026
Filtered objects, spectral sequences, and t-structures	January 2026
(A synthetic approach to detecting) v_1-periodic families	November 2025
Algebraic and Milnor K-theory	October 2025
Galois descent and the Picard group of K-theory	May 2025
The Adams spectral sequence and Hopf algebroids	February 2025
Symplectic orientations	August 2024